

In our case, if c is continuous and finite on a compact set, and hence uniformly continuous, this means that there exists an increasing continuous function $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $\omega(0) = 0$ such that

$$|c(x, y) - c(x', y')| \leq \omega(d(x, x') + d(y, y')).$$

Hence, when we take the definition of χ^c , we have $\chi^c(y) = \inf_x g_x(y)$ with $g_x(y) := c(x, y) - \chi(x)$ and the functions g_x satisfy $|g_x(y) - g_x(y')| \leq \omega(d(y, y'))$. This proves that χ^c shares the same continuity modulus also.

It is quite easy to realize that, given a pair (φ, ψ) in the maximization problem (DP), one can always replace it with (φ, φ^c) , and then with $(\varphi^{c\bar{c}}, \varphi^c)$, and the constraints are preserved and the integrals increased. Actually one could go on, but we will see that $\varphi^{c\bar{c}c} = \varphi^c$ for any φ (see Section 1.6). The goal of these transformations is to “improve” the maximizing sequence so as to get a uniform bound on its continuity.

A consequence of these considerations is the following existence result.

Proposition 1.11. *Suppose that X and Y are compact and c is continuous. Then there exists a solution (φ, ψ) to problem (DP) and it has the form $\varphi \in c\text{-conc}(X)$, $\psi \in \bar{c}\text{-conc}(Y)$, and $\psi = \varphi^c$. In particular*

$$\max(\text{DP}) = \max_{\varphi \in c\text{-conc}(X)} \int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu.$$

Proof. From the considerations above we can take a maximizing sequence (φ_n, ψ_n) and improve it, by means of c - and $\bar{c}$ -transforms, so that we can assume a uniform bound on the continuity of these functions (the same modulus of continuity as c). Instead of renaming the sequence, we will still call (φ_n, ψ_n) the new sequence obtained after these transforms. We only need to check equi-boundedness so as to apply Ascoli-Arzelà’s theorem. This may be done if we note that adding a constant to φ and subtracting it to ψ is always possible: the value of the functional does not change, nor the constraints are affected. Hence, since φ_n is continuous on a compact set and hence bounded, we can always subtract its minimum and suppose without loss of generality that $\min \varphi_n = 0$. We get $\max \varphi_n \leq \omega(\text{diam} X)$ (since the oscillation of a function is always less than its modulus of continuity computed at the highest possible distance in the set). So, if we have chosen $\psi_n = \varphi_n^c$, we also have $\psi_n(y) = \inf_x c(x, y) - \varphi_n(x) \in [\min c - \omega(\text{diam} X), \max c]$. This gives uniform bounds on φ_n and ψ_n and allows to apply Ascoli-Arzelà’s theorem.

Passing to a subsequence we can assume $\varphi_n \rightarrow \varphi$ and $\psi_n \rightarrow \psi$, both convergences being uniform. It is easy to see that

$$\int_X \varphi_n \, d\mu + \int_Y \psi_n \, d\nu \rightarrow \int_X \varphi \, d\mu + \int_Y \psi \, d\nu,$$

as a consequence of uniform convergence. Moreover

$$\varphi_n(x) + \psi_n(y) \leq c(x, y) \Rightarrow \varphi(x) + \psi(y) \leq c(x, y)$$

(here pointwise convergence would have been enough). This shows that (φ, ψ) is an admissible pair for (DP) and that it is optimal. $\square$ $\square$

If we admit the duality result $\min(\text{KP}) = \max(\text{DP})$ (the proof is postponed to Section 1.6), then we also have

$$\min(\text{KP}) = \max_{\varphi \in c\text{-conc}(X)} \int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu,$$

which also shows that the minimum value of (KP) is a convex function of (μ, ν) , as it is a supremum of linear functionals.

Definition 1.12. The functions φ realizing the maximum in (3.1) are called *Kantorovich potentials* for the transport from μ to ν . This is in fact a small abuse, because traditionally this term was used only in the case $c(x, y) = |x - y|$, but it is nowadays understood in the general case as well.

Remark 1.13. Note that Kantorovich potentials are not necessarily c -concave functions, but coincide with the c -concave function $\varphi^{c^c} \mu$ -a.e. However, it is always possible to choose a c -concave Kantorovich potential, and we will do it often.

1.3 The case $c(x, y) = h(x - y)$ for h strictly convex, and the existence of an optimal T

This section is mainly devoted to the results that we can obtain in the case where $X = Y = \Omega \subset \mathbb{R}^d$ and the cost c is of the form $c(x, y) = h(x - y)$, for a strictly convex function h . We will also assume Ω to be compact for simplicity. This case allows for very strong results, and in particular we will find existence, as well as a representation formula, for the optimal T. Anyway, the first few lines of this section will be concerned with a more general case: that of costs functions c satisfying a *twist condition*, the most remarkable case being exactly those of the form $h(x - y)$ with h strictly convex.

The main tool is the duality result. More precisely, we use Theorem 1.39, which applies to the case of compact domains with continuous cost functions c . It guarantees $\max(\text{DP}) = \min(\text{KP})$.

From the equality between the minimum of (KP) and the maximum of (DP) and the fact that both extremal values are realized, one can consider an optimal transport plan γ and a Kantorovich potential φ and write

$$\varphi(x) + \varphi^c(y) \leq c(x, y) \text{ on } \Omega \times \Omega \text{ and } \varphi(x) + \varphi^c(y) = c(x, y) \text{ on } \text{spt}(\gamma).$$

The equality on $\text{spt}(\gamma)$ is a consequence of the inequality which is valid everywhere and of

$$\int_{\Omega \times \Omega} c \, d\gamma = \int_{\Omega} \varphi \, d\mu + \int_{\Omega} \varphi^c \, d\nu = \int_{\Omega \times \Omega} (\varphi(x) + \varphi^c(y)) \, d\gamma,$$

which implies equality γ -a.e. These functions being continuous, the equality is satisfied on a closed set, i.e. on the support of the measure γ (let us recall the definition of support of a measure, not to be confused with sets where it is concentrated).

Definition 1.14. On a separable metric space X , the support of a measure γ is defined as the smallest closed set on which γ is concentrated, i.e.

$$\text{spt}(\gamma) := \bigcap \{A : A \text{ is closed and } \gamma(X \setminus A) = 0\}.$$

This is well defined since the intersection may be taken countable, due to the separability assumption. Moreover, there exists also this characterization

$$\text{spt}(\gamma) = \{x \in X : \gamma(B(x, r)) > 0 \text{ for all } r > 0\}.$$

Once we have that, let us fix a point $(x_0, y_0) \in \text{spt}(\gamma)$. One may deduce from the previous computations that

$$x \mapsto \varphi(x) - c(x, y_0) \quad \text{is minimal at } x = x_0$$

and, if φ and $c(\cdot, y_0)$ are differentiable at x_0 and $x_0 \notin \partial\Omega$, one gets $\nabla\varphi(x_0) = \nabla_x c(x_0, y_0)$. We resume this fact in a very short statement (where we do not put the sharpest assumptions on c) since we will use it much later on.

Proposition 1.15. *If c is C^1 , φ is a Kantorovich potential for the cost c in the transport from μ to ν , and (x_0, y_0) belongs to the support of an optimal transport plan γ , then $\nabla\varphi(x_0) = \nabla_x c(x_0, y_0)$, provided φ is differentiable at x_0 . In particular, the gradients of two different Kantorovich potentials coincide on every point $x_0 \in \text{spt}(\mu)$ where both the potentials are differentiable.*

Proof. The proof is contained in the above considerations. □ □

The equality $\nabla\varphi = \nabla_x c$ is particularly useful when c satisfies the following definition.

Definition 1.16. For $\Omega \subset \mathbb{R}^d$ we say that $c : \Omega \times \Omega \rightarrow \mathbb{R}$ satisfies the Twist condition whenever c is differentiable w.r.t. x at every point, and the map $y \mapsto \nabla_x c(x_0, y)$ is injective for every x_0 . This condition is also known in economics as Spence-Mirrlees condition (see, for instance, [261]). For “nice” domains and cost functions, it corresponds to $\det\left(\frac{\partial^2 c}{\partial y_i \partial x_j}\right) \neq 0$.

The goal of this condition is to deduce from $(x_0, y_0) \in \text{spt}(\gamma)$, that y_0 is indeed uniquely defined from x_0 . This shows that γ is concentrated on a graph, that of the map associating y_0 to each x_0 , and this map will be the optimal transport. Since this map has been constructed using φ and c only, and not γ , it also provides uniqueness for the optimal γ .

We will see this strategy with more details in the particular case where $c(x, y) = h(x - y)$, with h strictly convex, but the reader can see how to translate it into the most general case.

For this choice of c , if φ and h are differentiable at x_0 and $x_0 - y_0$, respectively, and $x_0 \notin \partial\Omega$, one gets $\nabla\varphi(x_0) = \nabla h(x_0 - y_0)$. This works if the function h is differentiable, if it is not we shall write $\nabla\varphi(x_0) \in \partial h(x_0 - y_0)$ (using the subdifferential of h , see

Box 1.12 in Section 1.6). For a strictly convex function h one may inverse the relation passing to $(\nabla h)^{-1}$ thus getting

$$x_0 - y_0 = (\nabla h)^{-1}(\nabla \varphi(x_0)).$$

Notice that the expression $(\nabla h)^{-1}$ makes sense for strictly convex functions h , independently of $h \in C^1$, thanks to the considerations on the invertibility of ∂h in Box 1.12.

This formula gives the solution to the transport problem with this cost, provided φ is differentiable a.e. with respect to μ . This is usually guaranteed by requiring μ to be absolutely continuous with respect to the Lebesgue measure, and using the fact that φ may be proven to be Lipschitz.

Box 1.9. – Memo – Differentiability of Lipschitz functions

Theorem (Rademacher) - Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ a Lipschitz continuous function. Then the set of points where f is not differentiable is negligible for the Lebesgue measure. We do not provide a proof of this fact, but one can look at Chapter 3 of [13].

Then, one may use the previous computation to deduce that, for every x_0 , the point y_0 (whenever it exists) such that $(x_0, y_0) \in \text{spt}(\gamma)$ is unique (i.e. γ is of the form $\gamma_T := (\text{id}, T)_\# \mu$ where $T(x_0) = y_0$). Moreover, this also gives uniqueness of the optimal transport plan and of the gradient of the Kantorovich potential.

We may summarize everything in the following theorem:

Theorem 1.17. *Given μ and ν probability measures on a compact domain $\Omega \subset \mathbb{R}^d$ there exists an optimal transport plan γ for the cost $c(x, y) = h(x - y)$ with h strictly convex. It is unique and of the form $(\text{id}, T)_\# \mu$, provided μ is absolutely continuous and $\partial\Omega$ is negligible. Moreover, there exists a Kantorovich potential φ , and T and the potentials φ are linked by*

$$T(x) = x - (\nabla h)^{-1}(\nabla \varphi(x)).$$

Proof. The previous theorems give the existence of an optimal γ and an optimal φ . The previous considerations show that if we take a point $(x_0, y_0) \in \text{spt}(\gamma)$ where $x_0 \notin \partial\Omega$ and $\nabla \varphi(x_0)$ exists, then necessarily we have $y_0 = x_0 - (\nabla h)^{-1}(\nabla \varphi(x_0))$. The points x_0 on the boundary are negligible by assumption. The points where the differentiability fails are Lebesgue-negligible by Rademacher's theorem. Indeed, φ shares the same modulus of continuity of c , which is a Lipschitz function on $\Omega \times \Omega$ since h is locally Lipschitz continuous and Ω is bounded. Hence, φ is also Lipschitz. From the absolute continuity assumption on μ , these two sets of “bad” points (the boundary and the non-differentiability points of φ) are μ -negligible as well. This shows at the same time that every optimal transport plan is induced by a transport map and that this transport map is $x \mapsto x - (\nabla h)^{-1}(\nabla \varphi(x))$. Hence, it is uniquely determined (since the potential φ does not depend on γ). As a consequence, we also have uniqueness of the optimal γ . □ □

Remark 1.18. All the costs of the form $c(x, y) = |x - y|^p$ with $p > 1$ can be dealt with via Theorem 1.17.

Remark 1.19. In the previous theorem we showed the uniqueness of the optimal plan by giving an explicit expression for the optimal map. Yet, it is possible to use a more general argument: every time that we know that any optimal γ must be induced by a map T , then we have uniqueness. Indeed, suppose that two different plans $\gamma_1 = \gamma_{T_1}$, $\gamma_2 = \gamma_{T_2}$ are optimal: consider $\frac{1}{2}\gamma_1 + \frac{1}{2}\gamma_2$, which is optimal as well by convexity. This last transport plan cannot be induced by a map unless $T_1 = T_2$ μ -a.e., which gives a contradiction.

Remark 1.20. Theorem 1.17 states that the optimal γ is of the form $(\text{id}, T)_\# \mu$, under some assumptions on μ . If the same assumptions are also satisfied by ν , then we can also say that there is an optimal map the other way around, i.e. $\gamma = (S, \text{id})_\# \nu$. In particular, γ -a.e. we have $y = T(x)$ and $x = S(y)$, which means $S(T(x)) = x$ γ -a.e., i.e. for μ -a.e. point x . Hence, T is invertible on a set of full measure and its inverse is the optimal map from ν to μ .

1.3.1 The quadratic case in $\mathbb{R}^d$

The case where the cost is given by $c(x, y) = \frac{1}{2}|x - y|^2$ in $\mathbb{R}^d$ deserves a special attention. The main reason for this special attention is the connection between c -concavity and the usual notion of convexity.

Proposition 1.21. *Given a function $\chi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{-\infty\}$ let us define $u_\chi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ through $u_\chi(x) = \frac{1}{2}|x|^2 - \chi(x)$. Then we have $u_{\chi^c} = (u_\chi)^*$. In particular, a function ζ is c -concave if and if $x \mapsto \frac{1}{2}|x|^2 - \zeta(x)$ is convex and l.s.c.*

Proof. Just compute

$$u_{\chi^c}(x) = \frac{1}{2}|x|^2 - \chi^c(x) = \sup_y \left(\frac{1}{2}|x|^2 - \frac{1}{2}|x - y|^2 + \chi(y) \right) = \sup_y x \cdot y - \left(\frac{1}{2}|y|^2 - \chi(y) \right).$$

This proves the first part of the statement. Moreover, since c -concave functions are characterized by the fact that they are c -transforms and convex l.s.c. functions by the fact that they are sup of affine functions (see the Box 1.11 in Section 1.6), the second part of the statement follows. $\square$ $\square$

As a consequence of the above proposition, we can particularize Theorem 1.17 to the quadratic case $c(x, y) = \frac{1}{2}|x - y|^2$, thus getting the existence of an optimal transport map

$$T(x) = x - \nabla \varphi(x) = \nabla \left(\frac{x^2}{2} - \varphi(x) \right) = \nabla u(x)$$

for a convex function u . Since we will also see the converse implication (sufficient optimality conditions), this will also prove the existence and the uniqueness of a gradient of a convex function transporting μ onto ν . This well known fact has been investigated first by Brenier (see [83]) and is usually referred to as Brenier Theorem. Section 1.7.2 will present the original approach by Brenier, called “polar factorization”.

Let us moreover note that a specific approach for the case $|x - y|^2$, based on the fact that we can withdraw the parts of the cost depending on x or y only and maximize $\int x \cdot y \, d\gamma$, gives the same result in a easier way: we actually get $\varphi(x_0) + \varphi^*(y_0) = x_0 \cdot y_0$ for a convex function φ and its Legendre transform φ^* , and we deduce $y_0 \in \partial\varphi(x_0)$ (see Box 1.12 in Section 1.6).

We note now that the existence of an optimal transport map is true under weaker assumptions on μ : we can replace the condition of being absolutely continuous with the condition “ $\mu(A) = 0$ for any $A \subset \mathbb{R}^d$ such that $\mathcal{H}^{d-1}(A) < +\infty$ ” or with any condition which ensures that the non-differentiability set of u is negligible.

In [5, 4] it is proven that the set where a convex function is not differentiable is $(d-1)$ -rectifiable (i.e. it is contained, up to $\mathcal{H}^{d-1}$ -negligible sets, in a countable union of Lipschitz images of $\mathbb{R}^{d-1}$); more generally, the set of points where the subdifferential has dimension at least k is $(d-k)$ -rectifiable. Lipschitz $(d-k)$ -surfaces can be replaced with C^1 surfaces, thanks to Lusin-type approximation results of Lipschitz functions via C^1 functions (see [161], chapter 6).

Also, it is possible to give an even stronger result about the non-differentiability set: it is contained $\mathcal{H}^{d-1}$ -a.e. in a countable union of $(d-1)$ -surfaces of class C^2 , as it is proven in [3]. Moreover, every set which is $(d-1)$ -rectifiable in this C^2 sense is contained in the non-differentiability set of a convex function³.

After these subtle details on the sharpest assumptions on μ which guarantee the existence of an optimal transport map T (essentially, the fact that convex functions must be differentiable μ -a.e.), we want now to adapt our analysis to the case of unbounded domains. This case is not covered by the general results of this section and has not been detailed so far. Yet, it can be handled replacing Theorem 1.39 with Theorem 1.40, which is valid for the cost $c(x, y) = \frac{1}{2}|x - y|^2$ without compactness assumptions on the domain (we just need $\int |x|^2 \, dx, \int |y|^2 \, dy < +\infty$). It provides the existence of solution (φ, ψ) to the following variant of (DP)

$$(\text{DP} - \text{var}) \quad \sup \left\{ \int_{\mathbb{R}^d} \varphi \, d\mu + \int_{\mathbb{R}^d} \psi \, d\nu : \varphi \in L^1(\mu), \psi \in L^1(\nu), \varphi \oplus \psi \leq c \right\},$$

together with $\max(\text{DP} - \text{var}) = \min(\text{KP})$. The sharpest result in the unbounded case is detailed in the following theorem.

Theorem 1.22. *Let μ, ν be probabilities over $\mathbb{R}^d$ and $c(x, y) = \frac{1}{2}|x - y|^2$. Suppose $\int |x|^2 \, dx, \int |y|^2 \, dy < +\infty$, which implies $\min(\text{KP}) < +\infty$ and suppose that μ gives no mass to $(d-1)$ -surfaces of class C^2 . Then there exists, unique, an optimal transport map T from μ to ν , and it is of the form $T = \nabla u$ for a convex function u .*

Proof. We follow again the same scheme as before: an optimal γ exists, and Theorem 1.40 also gives the existence of an optimal dual pair $(\varphi, \psi) \in L^1(\mu) \times L^1(\nu)$ and

³These considerations allowed N. Gigli in [183] to prove the following characterization theorem, extending 1.17: given $\mu, \nu \in \mathcal{P}(\Omega)$, suppose $\mu(A) = 0$ for every C^2 surface A of co-dimension 1; then the optimal transport plan $\gamma \in \Pi(\mu, \nu)$ for the quadratic cost $|x - y|^2$ is unique and induced by a transport map, whatever ν is. Moreover, the condition “ $\mu(A) = 0$ for every C^2 surface A of co-dimension 1” characterizes the class of measures μ such that the optimal transport plans (for the quadratic cost) from μ are unique and induced by maps for every target measure ν .

guarantees that we have no duality gap. Then, we can see that γ is concentrated on the graph of $x \mapsto x - \nabla \phi(x) := \nabla u(x)$ provided u (or ϕ , which is equivalent) is differentiable μ -a.e. Since $\phi \in L^1(\mu)$ we infer that u is finite μ -a.e. and, since u is convex, we get $\text{spt}(\mu) \subset \{u < +\infty\}$, which is a convex set. Note that $\partial(\{u < +\infty\})$ can be expressed locally as the graph of a concave function, hence it is $(d-1) - C^2$ rectifiable set, and is μ -negligible by assumption. In the interior of $\{u < +\infty\}$, u is differentiable μ -a.e. (again, because of the assumption on μ and of u being convex). $\square$ $\square$

Remark 1.23. As a consequence of all these considerations, the quadratic case gives a very interesting result in dimension one. Suppose that $\mu \in \mathcal{P}(\mathbb{R})$ is atomless. Then every convex function is differentiable μ -a.e., since we know that the set of non-differentiability points of a convex function is at most countable (this is a consequence of the fact that, if ψ is convex, then the intervals $]\psi'_l(x), \psi'_r(x)[$, where ψ'_l and ψ'_r denote the left and right derivatives, are all non-empty and disjoint when x ranges among non-differentiability points). This implies the existence of an optimal transport map for the quadratic cost between μ and any measure $\nu \in \mathcal{P}(\mathbb{R})$. This transport map will be the derivative of a convex function, i.e. a non-decreasing map.

Remark 1.24. Some of the properties of the quadratic case stay true if we pass to a general cost $c(x, y)$ which is $C^{1,1}$ and satisfies the twist condition of Definition 1.16. Obviously, the fact that the transport is the gradient of a convex function is no more true; on the other hand, the differentiability properties of Kantorovich potentials are the same as for convex functions, since one can prove that ϕ is semi-concave (i.e. concave, up to subtracting a quadratic function), and this allows to use the same assumptions on μ as in Theorem 1.22.

1.3.2 The quadratic case on the flat torus

It is useful and interesting to see what happens if we replace measures on $\mathbb{R}^d$ with measures on the flat torus $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ and the cost $\frac{1}{2}|x - y|^2$ with the squared distance on the torus, i.e. $c(x, y) = \frac{1}{2}||x - y||^2$, where

$$||z|| := \min\{|z + k| : k \in \mathbb{Z}^d\}. \quad (1.5)$$

This is the simplest example of optimal transport on a manifold, for which a general existence theorem has been established by McCann in [231]. Yet, the case of the torus, first studied in [128], is much simpler, because of the structure that $\mathbb{T}^d$ inherits from $\mathbb{R}^d$. In many cases (we will see in Chapters 5 and 6, in particular), studying what happens on the torus allows to understand the main qualitative features of the problem, getting rid of difficulties arising from lack of compactness and/or boundary issues. In the case of compactly supported measures, it is possible to view their supports as part of a large cube, to be identified with a torus, and this at no price. Indeed, if the measures are far from the boundary of the cube, the optimal transport maps on the torus and on the Euclidean space will coincide.

Note that an optimal $\bar{k}$ in (1.5) always exists. If we take $z \in Q := [-\frac{1}{2}, \frac{1}{2}]^d$, then $\bar{k} \in \{-1, 0, +1\}^d$, and it is unique (and equal to 0) unless $z \in \partial Q$. Moreover, the function $\mathbb{R}^d \ni z \mapsto \frac{1}{2}||z||^2$ is semi-concave, as it is the infimum of a family of functions

with Hessian equal to the identity matrix I . It is differentiable at every point where the optimal $\bar{k}$ is unique. For a given point x , we call *cutlocus* of x the set of points y such that the squared distance function, viewed as a function of the pair (x, y) , is not differentiable at (x, y) . This is a general notion in differential geometry, and we denote it by $\text{cut}(x)$ (see [294] for its use in optimal transport), but in the case of the flat torus it simply coincides with the set of points y such that $y - x \in \partial Q + \mathbb{Z}^d$.

Theorem 1.25. *Take $\mu, \nu \in \mathcal{P}(\mathbb{T}^d)$, with $\mu \ll \mathcal{L}^d$ and $c(x, y) = \frac{1}{2}||x - y||^2$. Then there exists a unique optimal transport plan $\gamma \in \Pi(\mu, \nu)$, it has the form $\gamma = \gamma_\Gamma$ and the optimal map T is given by $T(x) = x - \nabla \varphi(x)$ for a.e. x , where the sum $x - \nabla \varphi(x)$ is to be intended modulo $\mathbb{Z}^d$. Here the function φ is a Kantorovich potential, solution to the dual problem, in the transport from μ to ν for the cost c . Moreover, for a.e. $x \in \mathbb{T}^d$, the point $T(x)$ does not belong to $\text{cut}(x)$.*

Proof. The strategy is the same as for the costs of the form $h(x - y)$. We take an optimal transport plan γ , together with a pair (φ, ψ) , solution of the dual problem, and we fix $(x_0, y_0) \in \text{spt}(\gamma)$. We have

$$\frac{1}{2}||x - y_0||^2 - \varphi(x) \geq \psi(y_0), \quad \frac{1}{2}||x_0 - y_0||^2 - \varphi(x_0) = \psi(y_0).$$

Now, take $k_0 \in \mathbb{Z}^d$ such that $||x_0 - y_0|| = |x_0 - y_0 - k_0|$ and observe that, from the inequality $||x - y_0|| \leq |x - y_0 - k_0|$, we have

$$\frac{1}{2}|x - y_0 - k_0|^2 - \varphi(x) \geq \psi(y_0), \quad \frac{1}{2}|x_0 - y_0 - k_0|^2 - \varphi(x_0) = \psi(y_0).$$

This means that the function $x \mapsto \frac{1}{2}|x - y_0 - k_0|^2 - \varphi(x)$ is minimal at $x = x_0$. Yet, $\varphi(x) = \inf_y \frac{1}{2}||x - y||^2 - \psi(y)$ is a semi-concave function, as it is defined as an infimum of uniformly semi-concave functions. Hence, it is differentiable a.e. and μ -a.e.

If we assume that x_0 is such that $\nabla \varphi(x_0)$ exists, we deduce from the minimality above that $x_0 - y_0 - k_0 = \nabla \varphi(x_0)$. This implies that k_0 is uniquely determined by x_0 and y_0 , and hence $y_0 \notin \text{cut}(x_0)$. Moreover, y_0 is uniquely determined by x_0 and we get the usual result for the existence and uniqueness of the optimal map. $\square \quad \square$

Remark 1.26. In the proof of the previous theorem we can also observe that we have $k_0 = 0$ if $x_0 - y_0 \in Q$; this means in particular that whenever $\text{spt}(\mu), \text{spt}(\nu) \subset \frac{1}{2}Q$, then the transport is completely identical to what we would have with the classical quadratic cost in $\mathbb{R}^d$ instead of using the periodic framework of $\mathbb{T}^d$.

1.4 Counter-examples to existence

We want to give at least two examples that are not included in the statement of Theorem 1.17 and where an optimal transport does not exist. We concentrate on examples where the cost $c(x, y)$ would usually allow for existence results (take for instance $c(x, y) = |x - y|^2$), but the reason for this lack of existence is due to the measure μ . Other examples where this is due to the cost and not to the measures are possible, see for instance Ex(58) and Ex(59).